

Hadoop

This document describes the installation of *Hadoop* in standalone mode on the local *Windows* and *Linux* operating systems and its integration in the *IntelliJ* and *Eclipse* IDEs.

1 Local Installation

1.1 Java

Hadoop development requires a local installation of the *Java JDK* to run. Use **version 21** of the JDK. On Linux it is typically available in the package manager of your distribution, on Windows you have to download and install it manually. One of the free JDK implementations can be downloaded from <https://adoptium.net/temurin/releases/>. (Do **not** use Java 25 since it will block loading the Hadoop libraries by default on Windows)

Note: You often find so called "Java" preinstalled on computers. These installations are not only totally outdated, but they always contain the Java **JRE** and not the Java **JDK**. The JRE will not work with Hadoop, you have to use the JDK!

1.2 Download

The main Hadoop package is available from <https://hadoop.apache.org/>. Download the **binary version 3.4.1**. It has a size of approximately 1 GB, so the download might take a while to complete. (Do **not** download 3.4.2, 3.4.3 or 3.5.0; these versions currently do not work on Windows because of broken build dependencies)

1.3 Unpacking

Unpack the archive `hadoop-3.4.1.tar.gz` into a directory on your hard disk, for example, `C:\hadoop-3.4.1\` for Windows or `/opt/hadoop-3.4.1/` on Linux. Ensure the path name contains no space characters. You may use the extraction command

```
tar -xf archive.tar.gz -C /target/directory
```

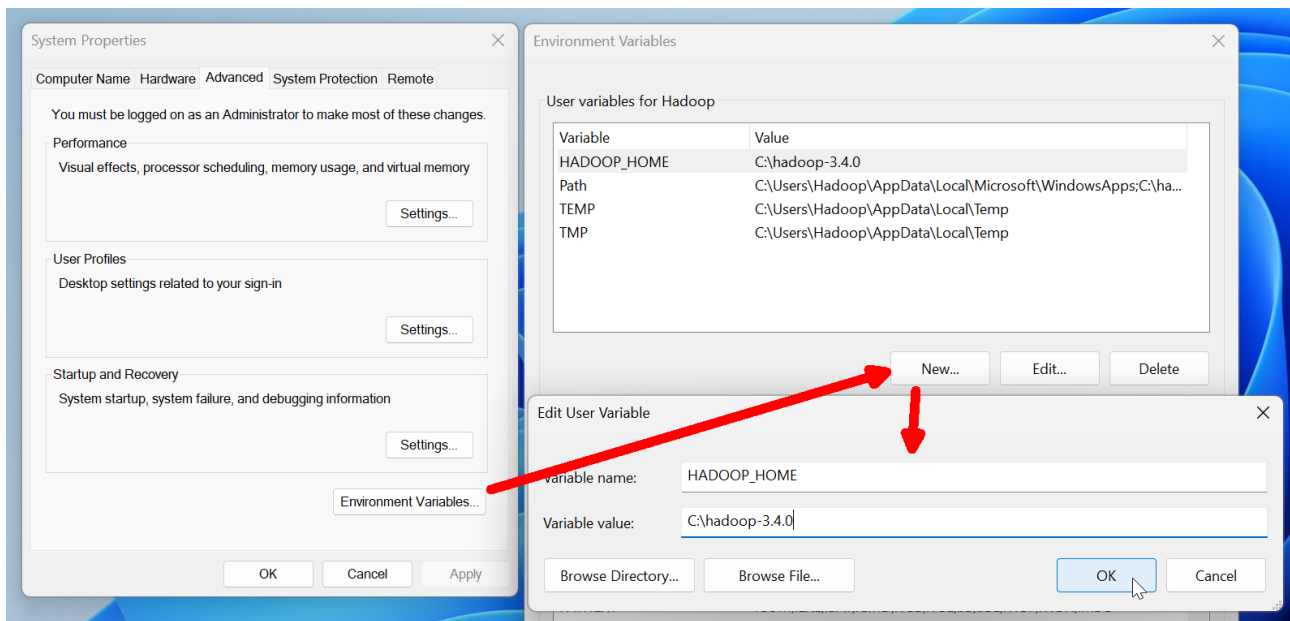
both on Windows and Linux. The archive `hadoop-3.4.1.tar.gz` contains symbolic file links, so you have to run the command with administrative permissions.

1.3.1 Additional Steps Windows

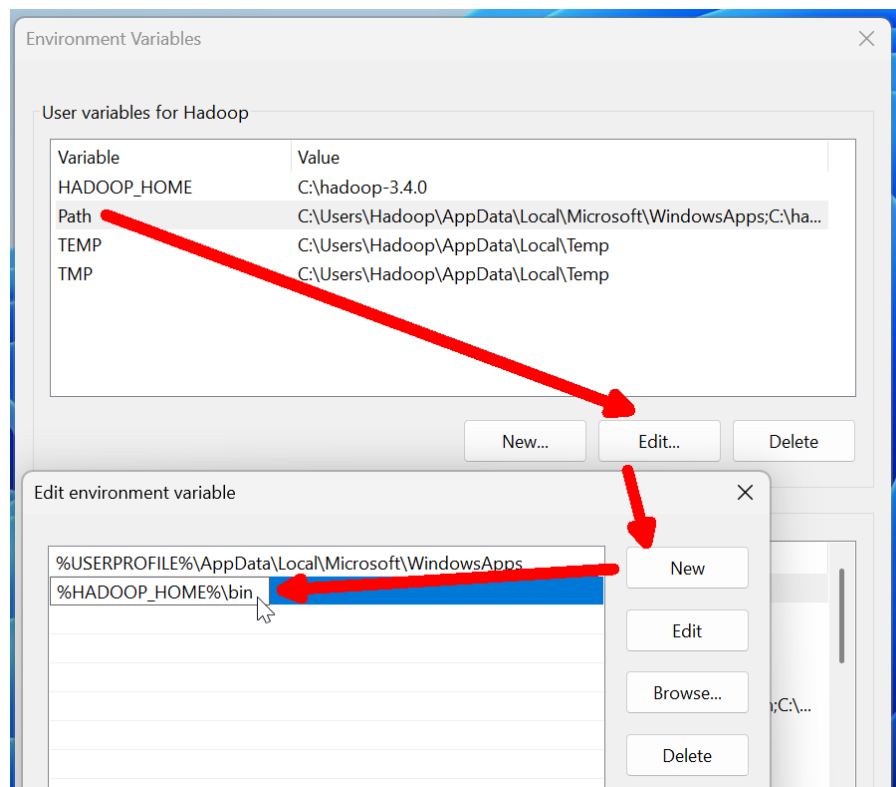
Running Hadoop on Windows requires additional libraries. On StudIP you can download the file `hadoop-3.4.1-windows-x64-dlls.zip`. Unpack its contents into your Hadoop directory.

After that, you have to modify your environment variables to allow Java to find the Hadoop libraries.

First, add a new variable named `HADOOP_HOME` and set it to your selected Hadoop installation folder. (**Note:** The screenshots are from version 3.4.0. Please use 3.4.1 in your setup)



Next, edit the existing variable `Path` and extend it with `%HADOOP_HOME%\bin`



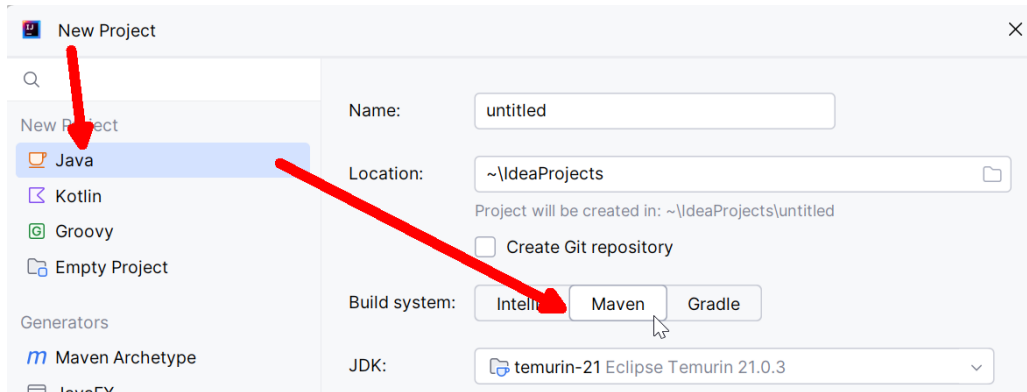
1.3.2 Additional Steps Linux

Similar to the Windows installation, you have to set the `HADOOP_HOME` environment variable to your Hadoop installation directory and extend the `PATH`. These entries are typically inserted into the `/etc/environment` file, but other locations are possible based on your distribution or desktop environment.

2 IDE Setup

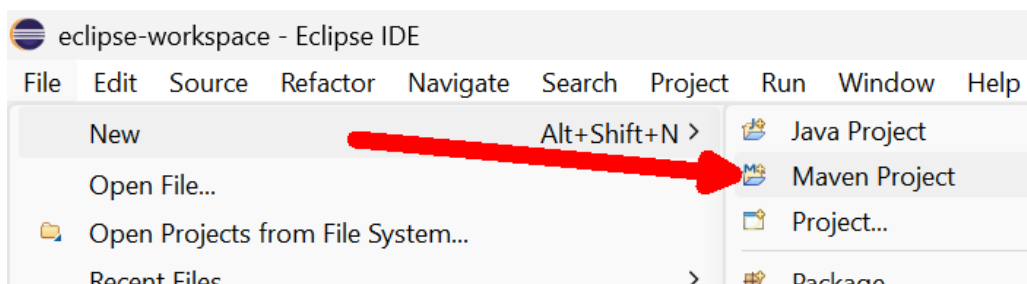
2.1 IntelliJ

Create a new *Maven* project:

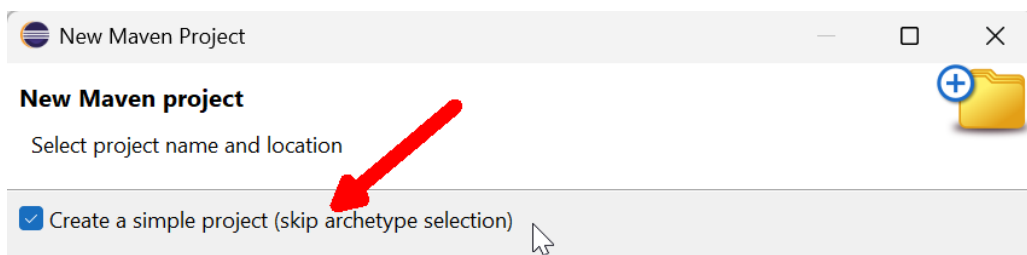


2.2 Eclipse

Create a new *Maven* project:



Create a simple project without archetype:



2.3 Both IDEs

Edit the newly created file `pom.xml` and insert the following dependencies inside the `<project>` XML Node:

```
<dependencies>
  <dependency>
    <groupId>org.apache.hadoop</groupId>
    <artifactId>hadoop-client-api</artifactId>
    <version>3.4.1</version>
  </dependency>
  <dependency>
    <groupId>org.apache.hadoop</groupId>
    <artifactId>hadoop-client-runtime</artifactId>
    <version>3.4.1</version>
  </dependency>
  <dependency>
    <groupId>org.slf4j</groupId>
    <artifactId>slf4j-simple</artifactId>
    <version>1.7.36</version>
  </dependency>
</dependencies>
```

The result should look similar to:



```
m pom.xml (Ex01-Maven) x
1 <?xml version="1.0" encoding="UTF-8"?>
2 <project xmlns="http://maven.apache.org/POM/4.0.0"
3       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
4       xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
5       http://maven.apache.org/maven-v4_0_0.xsd">
6   <modelVersion>4.0.0</modelVersion>
7   <artifactId>Ex01-Maven</artifactId>
8   <version>1.0-SNAPSHOT</version>
9   <properties>
10     <maven.compiler.source>21</maven.compiler.source>
11     <maven.compiler.target>21</maven.compiler.target>
12     <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
13   </properties>
14   <dependencies>
15     <dependency>
16       <groupId>org.apache.hadoop</groupId>
17       <artifactId>hadoop-client-api</artifactId>
18       <version>3.4.0</version>
19     </dependency>
20     <dependency>
21       <groupId>org.apache.hadoop</groupId>
22       <artifactId>hadoop-client-runtime</artifactId>
23       <version>3.4.0</version>
24     </dependency>
25     <dependency>
26       <groupId>org.slf4j</groupId>
27       <artifactId>slf4j-simple</artifactId>
28       <version>1.7.36</version>
29     </dependency>
30   </dependencies>
31 </project>
```

(it should show version 3.4.1 in your setup)

The first two dependencies are required for Hadoop, the third library is a simple logger which outputs Hadoop's status messages to the console. Without it, Hadoop will not write anything to the console, but runs correctly otherwise.